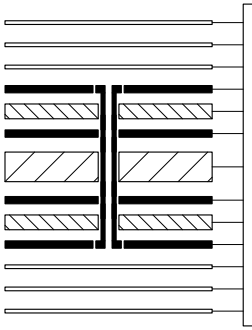
Dumb FCC Fabrication Document

Layer Stack Legend



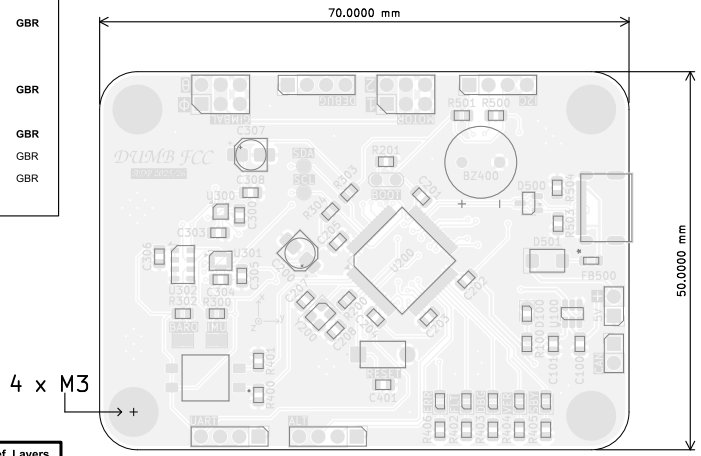
Material	Layer	Thickness	Dielectric	Type	Gerber
	F.Paste			Paste Mask	
	F.Silkscreen		Direct Printing	Legend	GBR
	F.Mask	0.02mm	Solder Resist	Solder Mask	GBR
Copper	**L1 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	**L2 (GND)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Core		1.15mm	FR4	Dielectric	
Copper	**L3 (GND)**	**0.035mm (1.00oz)**		**Plane**	**GBR**
Prepreg		0.1mm	FR4_7628	Dielectric	
Copper	**L4 (Sig, PWR)**	**0.07mm (2.00oz)**		**Signal**	**GBR**
	B.Mask	0.02mm	Solder Resist	Solder Mask	GBR
	B.Silkscreen		Direct Printing	Legend	GBR
	B.Paste			Paste Mask	

Total thickness: 1.6mm
Note: external layer thicknesses are specified after plating

Impedance Table

Transmission Line	Impedance [ohms]	Tolerance [ohms]	Layer	Trace Width [mm]	Gap [mm]	Ref. Layers
USB Differential Pair	90	+/- 10%	L1	0.1547	0.2	L2

Top Fabrication (Scale 1:1)



FABRICATION NOTES (UNLESS OTHERWISE SPECIFIED)

1) FABRICATE PER IPC-6012A CLASS 2.

2) OUTLINE DEFINED IN SEPARATE GERBER FILE WITH "Edge_Cuts.GBR" SUFFIX.

   DIMENSIONS OF CIRCUMSIZED RECTANGLE SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

3) SEE SEPARATE DRILL FILES WITH ".DRL" SUFFIX FOR HOLE LOCATIONS.

   SELECTED HOLE LOCATIONS SHOWN ON THIS DRAWING FOR REFERENCE ONLY.

4) SURFACE FINISH: HAL SNPB

5) SOLDERMASK ON BOTH SIDES OF THE BOARD SHALL BE LPI, COLOR PURPLE.

6) SILK SCREEN LEGEND TO BE APPLIED PER LAYER STACKUP USING WHITE NON-CONDUCTIVE EPOXY INK.

7) ALL VIAS ARE TENTED ON BOTH SIDES UNLESS SOLDERMASK OPENED IN GERBER.

8) VENDOR SHOULD FOLLOW ROHS COMPLIANT PROCESS AND Pb FREE FOR MANUFACTURING

9) PCB MATERIAL REQUIREMENTS:

   A. FLAMMABILITY RATING MUST MEET OR EXCEED UL94V-0 REQUIREMENTS.
   B. Tg 170 C OR EQUIVALENT.
   C. EQUIVALENT MATERIAL SHALL BE RoHS COMPLIANT, HALOGEN FREE AND APPROVED BY GROUP ALPHA.

10) DESIGN GEOMETRY MINIMUM FEATURE SIZES:

BOARD SIZE	70.000 × 50.000 mm
BOARD THICKNESS	1.600 mm
TRACE WIDTH	0.200 mm
TRACE TO TRACE	0.150 mm
MIN. HOLE (PTH)	0.300 mm
MIN. HOLE (NPTH)	0.600 mm
ANNULAR RING	0.195 mm
COPPER TO HOLE	0.300 mm
COPPER TO EDGE	0.500 mm
HOLE TO HOLE	0.250 mm

11) REFER TO IMPEDANCE TABLE FOR IMPEDANCE CONTROL REQUIREMENTS.

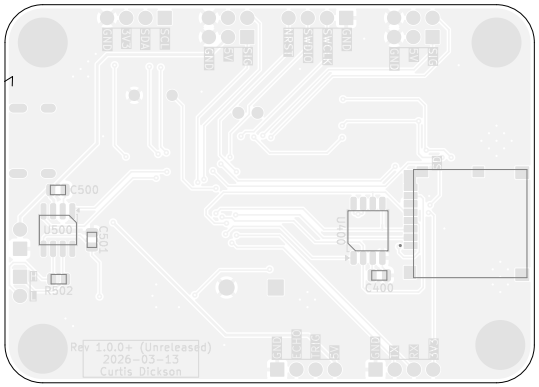
12) CONFIRM SPACE WIDTHS AND SPACINGS.

All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Group Alpha		Variant: PRELIMINARY	Git Hash: 261c7c2
	Board Name: **Dumb FCC**		Project Name: **ASDP4 2025/26**	
Sheet Title: Top Fabrication (Scale 1:1)	File Name: DumbFCC.kicad_pcb	Designer: Curtis Dickson	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **1** of **10**

Dumb FCC Fabrication Document

Bottom Fabrication (Scale 1:1)



All dimensions are in millimeters unless otherwise specified.

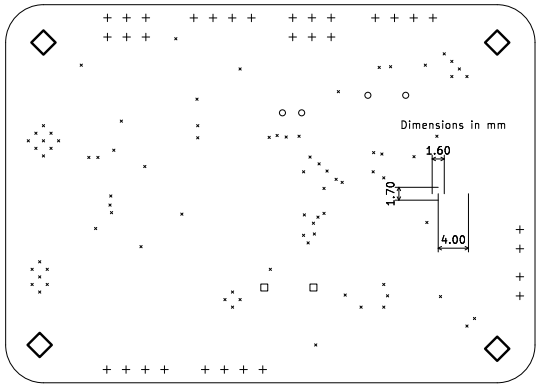
Comments:	Company: **Group Alpha**		Variant: PRELIMINARY	Git Hash: 261c7c2
	Board Name: **Dumb FCC**		Project Name: **ASDP4 2025/26**	
Sheet Title: Bottom Fabrication (Scale 1:1)	File Name: DumbFCC.kicad_pcb	Designer: Curtis Dickson	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: **A4**	Sheet: **2** of **10**

Dumb FCC Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
×	81	0.30mm (11.81mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Via
○	4	0.80mm (31.50mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
+	32	1.00mm (39.37mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
□	2	1.30mm (51.18mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
◇	4	3.20mm (125.98mils)	PTH	Round	L1 (Sig, PWR) - L4 (Sig, PWR)	Pad
	Total 123					

Drill Drawing L1 - L4 (Scale 1:1)



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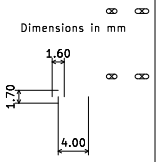
	Board Name: Dumb FCC		Project Name: ASDP4 2025/26	
Sheet Title: Drill Drawing (L1 - L4)	File Name: DumbFCC.kicad_pcb	Designer: Curtis Dickson	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 3 of 10

Dumb FCC Fabrication Document

Drill Table

Symbol	Count	Hole Size	Plated	Hole Shape	Drill Layer Pair	Hole Type
X	4	0.61mm (24.00mils)	NPTH	Slot	L1 (Sig, PWR) - L4 (Sig, PWR)	Mechanical
	Total 4					

Drill Drawing L1 - L4 (Scale 1:1)



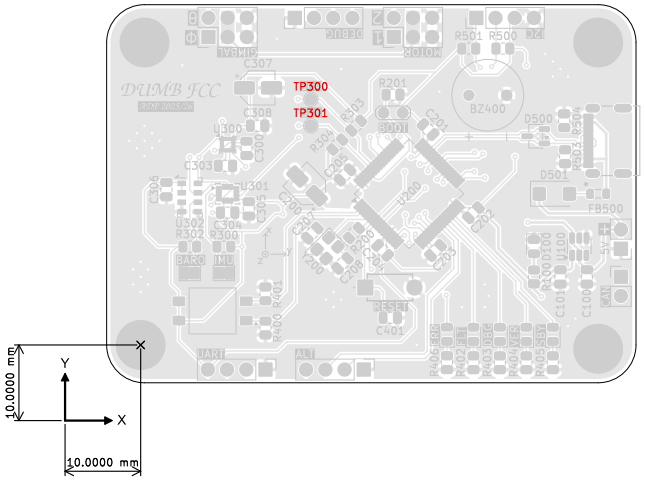
Comments:	Company: Group Alpha		Variant: PRELIMINARY	Git Hash: 261c7c2
	Board Name: Dumb FCC		Project Name: ASDP4 2025/26	
Sheet Title: Drill Drawing (L1 - L4)	File Name: DumbFCC.kicad_pcb	Designer: Curtis Dickson	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 4 of 10

Dumb FCC Fabrication Document

Top Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]
TP300	I2C_SENS_SDA	32.50	42.50
TP301	I2C_SENS_SCL	32.50	39.00

Ref.	Net	X [mm]	Y [mm]



All dimensions are in millimeters unless otherwise specified.

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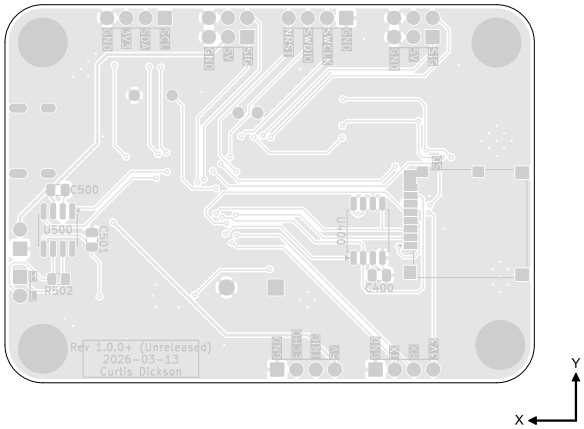
	Board Name: Dumb FCC		Project Name: ASDP4 2025/26	
Sheet Title: Top Test Points (Scale 1:1)	File Name: DumbFCC.kicad_pcb	Designer: Curtis Dickson	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 5 of 10

Dumb FCC Fabrication Document

Bottom Test Points (Scale 1:1)

Ref.	Net	X [mm]	Y [mm]



All dimensions are in millimeters unless otherwise specified.

Comments:	Company: Group Alpha		Variant: PRELIMINARY	Git Hash: 261c7c2
	Board Name: Dumb FCC		Project Name: ASDP4 2025/26	
Sheet Title: Bottom Test Points (Scale 1:1)	File Name: DumbFCC.kicad_pcb	Designer: Curtis Dickson	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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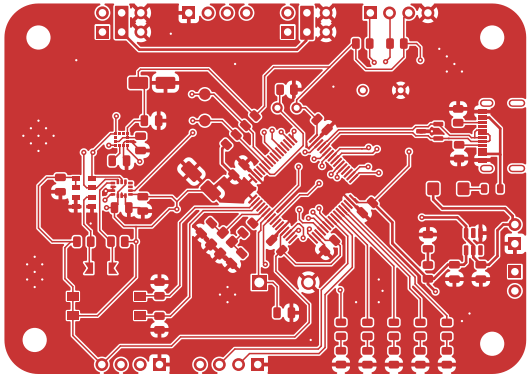
Dumb FCC Fabrication Document

L1 (Sig, PWR) (Scale 1:1)



Comments:	Company: Group Alpha		Variant: PRELIMINARY	Git Hash: 261c7c2
	Board Name: Dumb FCC		Project Name: ASDP4 2025/26	
Sheet Title: L1 (Sig, PWR) (Scale 1:1)	File Name: DumbFCC.kicad_pcb	Designer: Curtis Dickson	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 7 of 10

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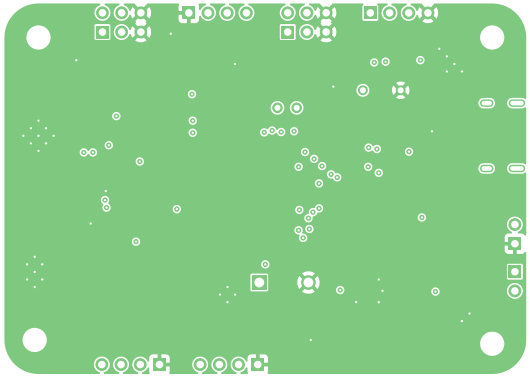
L2 (GND) (Scale 1:1)



Comments:	Company: Group Alpha		Variant: PRELIMINARY	Git Hash: 261c7c2
	Board Name: Dumb FCC		Project Name: ASDP4 2025/26	
Sheet Title: L2 (GND) (Scale 1:1)	File Name: DumbFCC.kicad_pcb	Designer: Curtis Dickson	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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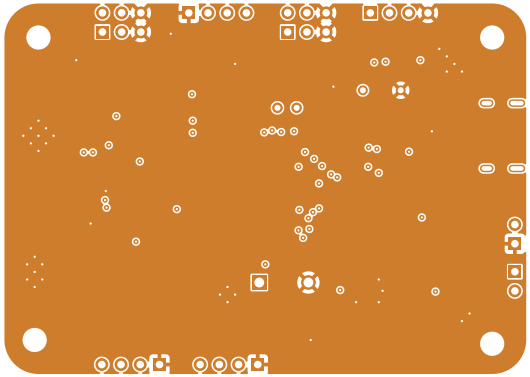
L3 (GND) (Scale 1:1)



Comments:	Company: Group Alpha		Variant: PRELIMINARY	Git Hash: 261c7c2
	Board Name: Dumb FCC		Project Name: ASDP4 2025/26	
Sheet Title: L3 (GND) (Scale 1:1)	File Name: DumbFCC.kicad_pcb	Designer: Curtis Dickson	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
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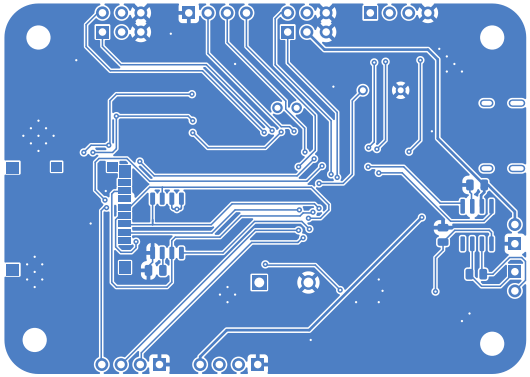
Dumb FCC Fabrication Document

L4 (Sig, PWR) (Scale 1:1)



Comments:	Company: Group Alpha		Variant: PRELIMINARY	Git Hash: 261c7c2
	Board Name: Dumb FCC		Project Name: ASDP4 2025/26	
Sheet Title: L4 (Sig, PWR) (Scale 1:1)	File Name: DumbFCC.kicad_pcb	Designer: Curtis Dickson	Date: 2024-04-13	Revision: 1.0.0+ (Unreleased)
Sheet Path:		Reviewer:	Size: A4	Sheet: 10 of 10